

TALAL NADEEM AWAN

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Education

COMSATS UNIVERSITY ISLAMABAD

Islamabad, PK

BS Cybersecurity, CGPA: 3.00

Relevant Coursework: Network Security, Artificial Intelligence, Information Security, Digital Forensics, Governance, Risk & Compliance (GRC), Web Development

WESTMINSTER ACADEMY

Islamabad, PK

A-Levels, 1A*, 1A

Experience

INARA TECHNOLOGIES

Islamabad, PK

Cybersecurity Intern

July 2025 – September 2025

- Built GitHub Actions CI/CD pipelines to automate security linting and vulnerability scanning for security-focused development projects.
- Conducted network-security assessments in lab environments using Nmap, Nessus, Ettercap, and Bettercap, identifying and troubleshooting security weaknesses across networked systems.
- Identified and tested vulnerabilities across OWASP Top 10 categories, including SQL injection, cross-site scripting (XSS), and access-control flaws using Burp Suite, PortSwigger Labs, and DVWA.
- Developed practical experience debugging technical issues, analyzing application and network behavior, and applying security principles to software-development workflows.

NATIONAL DATABASE & REGISTRATION AUTHORITY (NADRA)

Islamabad, PK

GRC Intern

July 2026 – August 2026

- Completed a six-week Governance, Risk & Compliance internship involving structured technical analysis, information-security risk assessment, auditing, and ISO/IEC 27001-aligned security practices.
- Conducted structured risk assessments by identifying organizational assets, analyzing associated risks, and documenting findings through detailed risk tables and risk registers.
- Completed a simulated startup exercise requiring the design and documentation of information-security and risk-management processes from the ground up.

- Performed an in-depth audit exercise and produced structured documentation of findings, risks, and recommended areas of treatment.

Projects

DocuMind-AI | Production-Ready Asynchronous RAG Engine

- Engineered an **asynchronous ingestion pipeline** using FastAPI, Redis, and Celery workers to decouple CPU-heavy document parsing and tokenization, maintaining non-blocking API response times (~20ms).
- Implemented **in-database vector search** utilizing PostgreSQL *pgvector* with Hierarchical Navigable Small World (**HNSW**) graph indexing, reducing nearest-neighbor search complexity to **$O(\log N)$** .
- Designed deterministic grounding safeguards, including a **pre-inference cosine similarity gate** (≥ 0.50 threshold) and **logit temperature scaling** ($T=0.2$) to eliminate hallucinations.
- Built an interactive **Provenance Inspector** exposing chunk UUIDs and **similarity scores** in real-time for verifiable, auditable source attribution.

Schrödinger's Detector | Python, PyTorch, PennyLane

- Developing a **hybrid quantum-classical neural network** combining a PyTorch CNN with PennyLane for **zero-day malware detection**.
- Designed the project around the application of **machine learning to cybersecurity** problems, experimenting with hybrid quantum and classical **neural-network architectures**.
- Documenting the research process, technical architecture, experimentation, and development through the Ares Vortex cybersecurity YouTube channel.

IronGate | Automated CI/CD Testing & Security Pipeline

- **Engineered automated testing and validation pipelines** in GitHub Actions, implementing deterministic pass/fail verification gates on every push and pull request.
- **Configured automated secret detection and regression testing**, benchmarking detection coverage across diverse token and credential formats against vulnerable web applications.
- **Debugged complex runner environment and dependency isolation issues**, resolving C-extension compilation errors and OS-level runtime constraints.
- **Structured reproducible execution harnesses** to support extensible verification checks across distributed workflows.

ThreatLens | Digital Asset Intelligence & Monitoring Dashboard

- **Built and deployed a responsive, full-stack intelligence dashboard** utilizing Next.js, TypeScript, and Vercel for monitoring digital attack surfaces.
- **Implemented modular, reusable UI and data-fetching components** to handle asynchronous API payloads with minimal client-side re-renders.
- **Designed structured state management and error-handling workflows**, ensuring reliable data visualization across diverse asset query endpoints.

LawyerJr | Paid Client Legal Study Chatbot

- Built and delivered LawyerJr as a paid chatbot project designed to assist a law student with legal studies through a conversational interface.
- Designed the chatbot around a real client use case, focusing on making interaction with legal study material more accessible and practical.
- Translated the client's academic requirements into a functional software product, gaining experience delivering a custom chatbot for an external user.

Skills & Interests

Programming: Python, JavaScript, TypeScript, Java, C++

AI & Machine Learning: PyTorch, PennyLane, Neural Networks, Machine Learning, Chatbot Development

Frontend: React, Next.js, HTML, CSS

Backend & API's: Node.js, FastAPI, REST API Concepts

Databases: PostgreSQL, MongoDB

DevOps & Cloud: Git, GitHub Actions, CI/CD, Docker, Vercel, AWS Basics

Cybersecurity: Application Security, Network Security, OWASP Top 10, Burp Suite, Nmap, Nessus, PortSwigger Labs

Soft Skills: Team Leading, Team Motivation, Negotiation, Public Speaking, Adaptability, Customer Service, Problem Solving, Team Player

Language: English (native-level fluency)

Interests: Football, Athletics, Calisthenics, Video editing, Video Games

Certifications

ISC2 Certifications:

- Certified in Cybersecurity (CC) — In Progress

OPSWAT Certifications:

- Introduction to Critical Infrastructure Protection (CIP)